Embedded Security Lock System

Using Arduino Microcontroller

A Project Report Submitted in Partial Fulfillment
of the Requirements for the Course

Embedded Systems Design

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ABSTRACT

This project presents the design and implementation of an embedded security lock system using Arduino Uno microcontroller. The system provides dual-layer authentication by combining RFID (Radio Frequency Identification) technology with a 4-digit password system.

When a user scans a valid RFID tag, the system prompts for a password. Upon correct password entry, a servo motor unlocks the door. The system provides real-time feedback through an LCD display, LED indicators, and a buzzer. This cost-effective solution is suitable for homes, offices, and industrial applications.

Keywords: Arduino Uno, RFID, Security System, Password Authentication, Access Control, Servo Motor

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1 INTRODUCTION

1.1 Background

In today's technology-driven world, security has become a critical concern. Traditional mechanical locks are vulnerable to picking, key duplication, and forced entry. Electronic security systems offer enhanced protection through features like RFID authentication, password protection, and biometric verification.

This project addresses these security gaps by developing an affordable, reliable, and easy-to-use electronic lock system.

1.2 Project Objective

The main objectives of this project are:

- Design a dual-factor authentication system (RFID + Password)
- Implement real-time user feedback using LCD and LEDs
- Create an automatic locking mechanism using a servo motor
- Ensure the system is cost-effective and easy to install

1.3 Applications

This security system can be used in:

- **Smart Homes** - Door access control
- **Offices** - Restricted area access
- **Laboratories** - Equipment security
- **Hostels/Hotels** - Room access management
- **Industrial Control** - Machine access control
- **Healthcare** - Medicine cabinet security

1.4 Report Organization

This report is organized into six sections:

1. Introduction - Background and objectives
2. Literature Review - Related work and technology overview
3. System Design - Hardware and software architecture
4. Implementation - Step-by-step setup and code
5. Results - Testing and performance analysis
6. Conclusion - Summary and future enhancements

2 LITERATURE REVIEW

2.1 Existing Security Systems

Various security technologies exist in the market. Table 1 compares different approaches.

Table 1: Comparison of Security Technologies

Technology	**Cost**	**Security Level**	**Complexity**
Mechanical Lock	Low	Low	Simple
Keypad Lock	Medium	Medium	Moderate
RFID System	Medium	Medium-High	Moderate
Biometric System	High	High	Complex
Our System (RFID+Password)	**Medium**	**High**	**Moderate**

2.2 RFID Technology

RFID (Radio Frequency Identification) uses electromagnetic fields to identify tags attached to objects. The MFRC522 module used in this project:

- Operating frequency: 13.56 MHz
- Read range: 3-5 cm
- Supports both tags and cards
- Communicates via SPI protocol

2.3 Arduino Platform

Arduino Uno is an open-source microcontroller board based on the ATmega328P. Key specifications:

- 14 digital input/output pins
- 6 analog inputs
- 16 MHz quartz crystal
- USB connection for programming
- Operating voltage: 5V
- Input voltage: 7-12V

3 SYSTEM DESIGN

3.1 Block Diagram

Input Devices	Controller	Output Devices
RFID Reader →	Arduino Uno	→ Servo Motor
Keypad →	(ATmega328P)	→ LCD Display
		→ Green LED
		→ Red LED
		→ Buzzer

Figure 1: Simplified Block Diagram

3.2 Components Required

Table 2: Components List with Specifications

Component	**Specification**	**Qty**
Arduino Uno	ATmega328P	1
RFID Reader	MFRC522 (13.56 MHz)	1
RFID Tags/Cards	Passive 13.56 MHz	2
4x4 Matrix Keypad	Membrane type	1
LCD Display	16x2 with I2C module	1
Servo Motor	SG90	1
Red LED	5mm	1
Green LED	5mm	1
Buzzer	5V Active	1
Resistors	220Ω	2
Breadboard	830 points	1
Jumper Wires	Male-Female / Male-Male	30
Power Supply	7-12V DC	1

3.3 Pin Connections

Table 3: Pin Configuration Table

Component	**Arduino Pin**	**Purpose**
RFID - SS	Pin 10	Slave Select
RFID - RST	Pin 9	Reset
Servo Motor	Pin 8	PWM Control
Green LED (+)	Pin 7	Access Granted
Red LED (+)	Pin 6	Access Denied
Buzzer (+)	Pin 5	Audio Alert
Keypad Rows	A0, A1, A2, A3	Row Inputs
Keypad Columns	Pins 2, 1, 0	Column Outputs
LCD - SDA	A4	I2C Data
LCD - SCL	A5	I2C Clock
All GNDs	GND	Common Ground
All 5V	5V	Power Supply

3.4 Working Principle

The system operates in two authentication stages:

3.4.1 Stage 1: RFID Authentication

1. LCD displays "Scan Your Tag"
2. User presents RFID tag to the reader
3. System reads the tag's unique UID
4. If UID matches stored value → Proceed to Stage 2
5. If UID does NOT match → Red LED + Buzzer → Access Denied

3.4.2 Stage 2: Password Authentication

1. LCD displays "Enter Password:"
2. User enters 4-digit password on keypad
3. System compares with stored password
4. If correct → Green LED ON + Servo rotates 90° (Unlock)
5. After 3 seconds → Servo returns to 0° (Lock)
6. If incorrect → Red LED + Buzzer → Access Denied

4 IMPLEMENTATION

4.1 Hardware Setup Instructions

Step 1: Power Connections

- Connect Arduino to power source (7-12V via barrel jack or USB)
- Connect breadboard power rails to Arduino 5V and GND

Step 2: LCD Connection (I2C)

- LCD VCC → Arduino 5V
- LCD GND → Arduino GND
- LCD SDA → Arduino A4
- LCD SCL → Arduino A5

Step 3: Keypad Connection

- Row pins (R1-R4) → Arduino A0, A1, A2, A3
- Column pins (C1-C4) → Arduino Pin 2, 1, 0

Step 4: Servo Motor Connection

- Red wire (VCC) → Arduino 5V
- Brown/Black wire (GND) → Arduino GND
- Orange/Yellow wire (Signal) → Arduino Pin 8

Step 5: LED Connections

- Green LED (+) → 220 resistor → Arduino Pin 7
- Red LED (+) → 220 resistor → Arduino Pin 6
- Both LED (-) → Arduino GND

Step 6: Buzzer Connection

- Buzzer (+) → Arduino Pin 5
- Buzzer (-) → Arduino GND

Step 7: RFID Reader Connection

- SDA → Pin 10
- SCK → Pin 13
- MOSI → Pin 11
- MISO → Pin 12
- RST → Pin 9
- 3.3V → 3.3V (Arduino)
- GND → GND

4.2 Software Setup

Required Libraries:

1. Open Arduino IDE
2. Go to **Sketch** → **Include Library** → **Manage Libraries**
3. Install these libraries:
   - `MFRC522` by Miguel Balboa
   - `LiquidCrystal I2C` by Frank de Brabander
   - `Keypad` by Mark Stanley

4.3 Complete Arduino Code

```
/*
   EMBEDDED SECURITY LOCK SYSTEM
   Author: Srujitha
   Roll No: BT23ECI042

   This code implements a dual-factor authentication system
   using RFID tag scanning followed by password entry.
*/

#include <SPI.h>            // SPI communication for RFID
#include <MFRC522.h>        // RFID module library
#include <LiquidCrystal_I2C.h>  // LCD with I2C
#include <Keypad.h>         // Keypad library
#include <Servo.h>          // Servo motor library

// ========== PIN DEFINITIONS ==========
#define SS_PIN    10        // RFID Slave Select
#define RST_PIN   9         // RFID Reset
#define SERVO_PIN 8         // Servo motor signal
#define GREEN_LED 7         // Green LED (Access granted)
#define RED_LED   6         // Red LED (Access denied)
#define BUZZER    5         // Buzzer

// ========== INITIALIZE COMPONENTS ==========
LiquidCrystal_I2C lcd(0x27, 16, 2);  // I2C address 0x27
MFRC522 rfid(SS_PIN, RST_PIN);
Servo doorLock;

// ========== KEYPAD SETUP ==========
const byte ROWS = 4;
const byte COLS = 4;
char keys[ROWS][COLS] = {
    {'1','2','3','A'},
    {'4','5','6','B'},
    {'7','8','9','C'},
```

```
    {'*','0','#','D'}
};
byte rowPins[ROWS] = {A0, A1, A2, A3};
byte colPins[COLS] = {2, 1, 0};
Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS,
   COLS);

// ========== VARIABLES ==========
String validUID = "29 B9 ED 23";    // Your RFID tag's UID
char correctPassword[4] = {'1','2','3','4'};  // Default password
char enteredPassword[4];
int passwordIndex = 0;
bool rfidMode = true;    // true = waiting for RFID, false =
   waiting for password

// ========== SETUP FUNCTION ==========
void setup() {
    // Initialize pins
    pinMode(GREEN_LED, OUTPUT);
    pinMode(RED_LED, OUTPUT);
    pinMode(BUZZER, OUTPUT);

    // Initialize servo
    doorLock.attach(SERVO_PIN);
    doorLock.write(0);    // Start in locked position (0 degrees)

    // Initialize LCD
    lcd.init();
    lcd.backlight();
    lcd.clear();

    // Initialize RFID
    SPI.begin();
    rfid.PCD_Init();
}

// ========== MAIN LOOP ==========
void loop() {
    if (rfidMode) {
        // MODE 1: Wait for RFID tag
        lcd.setCursor(0, 0);
        lcd.print("  Door Lock     ");
        lcd.setCursor(0, 1);
        lcd.print(" Scan Your Tag ");

        // Check for RFID tag
        if (!rfid.PICC_IsNewCardPresent()) {
            return;
        }
        if (!rfid.PICC_ReadCardSerial()) {
            return;
```

```
        }

        // Read UID from tag
        String tagUID = "";
        for (byte i = 0; i < rfid.uid.size; i++) {
            tagUID += String(rfid.uid.uidByte[i], HEX);
            if (i < rfid.uid.size - 1) tagUID += " ";
        }
        tagUID.toUpperCase();

        // Verify UID
        if (tagUID == validUID) {
            // VALID TAG
            lcd.clear();
            lcd.print("  Tag Matched  ");
            digitalWrite(GREEN_LED, HIGH);
            delay(2000);
            digitalWrite(GREEN_LED, LOW);

            lcd.clear();
            lcd.print(" Enter Password:");
            lcd.setCursor(0, 1);
            rfidMode = false;    // Switch to password mode
        } else {
            // INVALID TAG
            lcd.clear();
            lcd.print(" Wrong Tag!    ");
            lcd.setCursor(0, 1);
            lcd.print(" Access Denied ");
            digitalWrite(RED_LED, HIGH);
            digitalWrite(BUZZER, HIGH);
            delay(3000);
            digitalWrite(RED_LED, LOW);
            digitalWrite(BUZZER, LOW);
            lcd.clear();
        }

        rfid.PICC_HaltA();

    } else {
        // MODE 2: Wait for password
        char key = keypad.getKey();

        if (key) {
            enteredPassword[passwordIndex++] = key;
            lcd.print("*");    // Show * for each entered digit
        }

        if (passwordIndex == 4) {
            delay(200);

```

```
            // Verify password
            if (enteredPassword[0] == correctPassword[0] &&
                enteredPassword[1] == correctPassword[1] &&
                enteredPassword[2] == correctPassword[2] &&
                enteredPassword[3] == correctPassword[3]) {

                // CORRECT PASSWORD - UNLOCK DOOR
                lcd.clear();
                lcd.print("   ACCESS      ");
                lcd.setCursor(0, 1);
                lcd.print("   GRANTED!    ");

                digitalWrite(GREEN_LED, HIGH);
                doorLock.write(90);    // Unlock (rotate 90 degrees)
                delay(3000);
                doorLock.write(0);     // Lock (return to 0 degrees)
                digitalWrite(GREEN_LED, LOW);

            } else {
                // WRONG PASSWORD
                lcd.clear();
                lcd.print(" WRONG PASSWORD");
                lcd.setCursor(0, 1);
                lcd.print(" Access Denied ");

                digitalWrite(RED_LED, HIGH);
                digitalWrite(BUZZER, HIGH);
                delay(3000);
                digitalWrite(RED_LED, LOW);
                digitalWrite(BUZZER, LOW);
            }

            // Reset for next attempt
            lcd.clear();
            passwordIndex = 0;
            rfidMode = true;
        }
    }
}
```

Listing 1: Full Arduino Code for Security Lock System

4.4 How to Find Your RFID Tag's UID

Upload this simple code to find your tag's UID:

```
#include <SPI.h>
#include <MFRC522.h>

#define SS_PIN 10
```

```
#define RST_PIN 9

MFRC522 rfid(SS_PIN, RST_PIN);

void setup() {
    Serial.begin(9600);
    SPI.begin();
    rfid.PCD_Init();
    Serial.println("Scan an RFID tag...");
}

void loop() {
    if (rfid.PICC_IsNewCardPresent() && rfid.PICC_ReadCardSerial
       ()) {
        Serial.print("UID: ");
        for (byte i = 0; i < rfid.uid.size; i++) {
            Serial.print(rfid.uid.uidByte[i], HEX);
            Serial.print(" ");
        }
        Serial.println();
        rfid.PICC_HaltA();
    }
}
```

Listing 2: RFID UID Reader Sketch

5 RESULTS AND TESTING

5.1 Test Scenarios

Table 4: Test Results Summary

Test Case	Expected Behavior	Result
Valid RFID + Valid Password (1234)	Door unlocks, Green LED ON	PASS
Valid RFID + Wrong Password	Access denied, Red LED + Buzzer	PASS
Invalid RFID Tag	Immediate access denied	PASS
No RFID tag present	System continues scanning	PASS
Power cycle (reset)	System returns to RFID mode	PASS

5.2 Performance Metrics

Table 5: Performance Analysis

Parameter	Value
RFID detection time	100-200 ms
Password verification time	¡50 ms
Servo unlock time	500 ms
Total authentication time	2-3 seconds
Idle current draw	150 mA
Active current draw	200 mA

5.3 Project Cost

Table 6: Cost Breakdown (Indian Rupees)

Component	Cost ()
Arduino Uno	1,500
RFID Kit (Reader + 2 tags)	800
LCD 16x2 with I2C	400
4x4 Keypad	200
Servo Motor SG90	350
Buzzer, LEDs, Resistors	100
Breadboard + Wires	250
Total	**3,600**

6 CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The embedded security lock system was successfully designed and implemented using Arduino Uno. The project achieved all its objectives:

- Dual-factor authentication (RFID + Password)
- Real-time feedback using LCD, LEDs, and buzzer
- Automatic locking/unlocking via servo motor
- Cost-effective solution under 4,000

The system provides reliable security for doors, cabinets, and access points. Its modular design allows for easy upgrades and customization.

6.2 Future Enhancements

The following features can be added to improve the system:

1. **WiFi Connectivity** - Add ESP8266 module for remote monitoring and control via smartphone
2. **Fingerprint Sensor** - Add biometric authentication for three-factor security
3. **Multiple Users** - Store multiple RFID tags and passwords for different users
4. **Access Log** - Record date/time of each access attempt on SD card
5. **Mobile App** - Create Android app for remote unlocking and monitoring
6. **Time Scheduling** - Allow access only during specific hours
7. **Email/SMS Alerts** - Send notifications for unauthorized access attempts

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APPENDIX

A. Troubleshooting Guide

Problem	Solution
LCD shows nothing	Check I2C connections. Try address 0x3F instead of 0x27. Adjust contrast pot.
RFID not reading tags	Check 3.3V power. Verify SPI pins. Hold tag closer to antenna.
Servo not moving	Check 5V power supply. Verify signal pin. Test servo separately.
Keypad not working	Verify row/column pin mapping. Check for stuck keys.
Buzzer/LED not working	Check polarity. Verify pin numbers in code.

B. How to Change Default Password

In the code, find this line:

```
char correctPassword[4] = {'1','2','3','4'};
```

Change the numbers to your desired 4-digit password.

C. How to Register a New RFID Tag

1. Upload the "RFID UID Reader" sketch (provided above)
2. Open Serial Monitor (9600 baud)
3. Scan your new tag
4. Copy the displayed UID
5. Replace the `validUID` value in main code

D. Bill of Materials (Detailed)

Component	Model/Part	Qty	Approx Price ()
Arduino Uno	R3 (ATmega328P)	1	1,500
RFID Module	MFRC522	1	600
RFID Tags (white card)	13.56 MHz	2	200
Keypad	4x4 Matrix Membrane	1	200
LCD Module	1602A with I2C	1	400
Servo Motor	SG90 Tower Pro	1	350
LED (Green)	5mm, 3V	1	10
LED (Red)	5mm, 3V	1	10
Buzzer	5V Active	1	30
Resistor	220Ω, 1/4W	2	10
Breadboard	830 tie-points	1	200
Jumper Wires	Male-Female (40pcs)	1	150

Power Supply	12V, 1A (optional)	1	200
USB Cable	A to B	1	100